

SAYANI MALICK

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EDUCATION

MS, Computer Science

University of Texas, Dallas

Richardson, USA

Aug 2025 - May 2027

4.0/4.0 GPA

Relevant coursework: Artificial Intelligence, Database Design, Computer Vision, Natural Language Processing

B.E., Electronics and Communication Engineering

BITS Pilani, KK Birla Goa Campus

Goa, India

Aug 2019 - Jul 2023

8.3 GPA

Relevant coursework: DSA, Object Oriented Programming, Neural Networks, Probability and Statistics

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C++, S

Data & Backend: SQL, MongoDB, REST APIs, JSON, Kafka, Spark, Hadoop, Big Data, Elasticsearch, Docker

AI / ML: Machine Learning, Matplotlib, PyTorch, TensorFlow, Scikit-learn, Image Processing

AI & Agents: Prompt engineering, LLMs, retrieval-style pipelines, LangChain, API integration

Tools: Git/GitHub, Jenkins, CI/CD, Linux, Shell scripting, Jest, Pytest, A/B testing

EXPERIENCE

Flipkart, Bangalore: Software Development Engineer - I and intern

Jul 2023 - Aug 2025

- Built Morpheus UI and backend pipelines for Flipkart's storefront using Spark, Elasticsearch, SQL, and Java REST APIs, reducing manual work by 80 hours/month and improving deployment efficiency by 50% in production.
- Designed a priority-based ingestion scheduler with MongoDB and Kafka Streams, cutting product refresh latency from 12 hours to 2 hours and boosting throughput and SLA compliance.
- Engineered a Kafka-Spark-Hadoop (MapReduce + Java) pipeline to compute new vs. repeat customer metrics, powering the Brand Insights Tool and driving a 20% increase in new brand listings.

Eclipse Adoptium, Remote: Software Development Intern [Link]

Jun 2021 - September 2021

- Redesigned the external testing framework using Java, XML, Shell scripting, and Jenkins, eliminating **90%** of redundant code to add new tests in the framework and introducing generic, reusable testing strategies, which streamlined inclusion of tests for Java test-suites.

PROJECTS

Machine Learning-Led Closed Loop Adaptation of Exergaming – HeXD Lab, UTD

Designed a deep-learning powered exergame agent that adapts dance difficulty in **real time** by integrating ECG signals, computer vision pose estimation, and game state under the guidance of Prof. Rawan Alghofaili.

Built Python data pipelines to ingest and preprocess physiological and pose data and trained ML models to estimate exertion and fatigue, driving automated difficulty adjustments in a closed-loop, agent-style control system.

TrustIssues.AI – Multi-Agent Compliance & Onboarding Assistant (HackUTD 2025 Winner)

Developed a Nemotron-powered multi-agent system that automates vendor/client onboarding by orchestrating document parsing, identity verification, deterministic risk scoring, and human-in-the-loop approvals.

Sleep apnea detection system using machine learning on resource-constrained devices

Developed self-contained systems using TinyML models and TensorFlow Lite deployed on resource-constrained devices like Raspberry Pi Pico and ESP32 to detect Sleep Apnea from ECG signals in real-time.

Novel Deep Learning Framework for Detection of Epileptic Seizures [Paper]

Developed novel 1D-CNN + Bi-LSTM architecture for epileptic seizure detection, achieving 98.03% accuracy on the Bonn EEG dataset; demonstrated expertise in time-series modeling and deep learning for biomedical signals

ACHIEVEMENTS

Winner, HackUTD 2025: Goldman Sachs Challenge: Winner at HackUTD 2025 for the Goldman Sachs Challenge, developing TrustIssues.ai: a Nemotron-powered multi-agent system to streamline vendor/client onboarding.

Research Week With Google, 2023: Selected as a scholar to collaborate with eminent scientists at Google Research and leading academic institutions, gaining exposure to cutting-edge advances in AI and computer science.